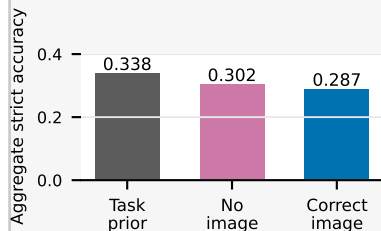


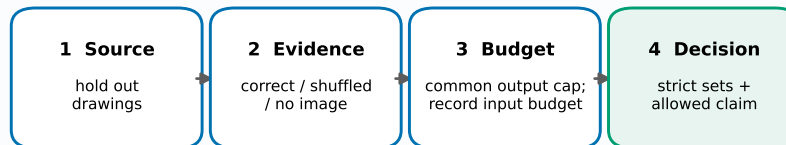
A qualification decision for source-conditioned P&ID tag retrieval

Aggregate accuracy is a diagnostic boundary; a retrieval claim must pass source, evidence, budget, and scope gates.

Why aggregate accuracy cannot qualify reading



SABER-PID qualification workflow



Pass only when requested-drawing and matched-budget contrasts support the same task-specific interpretation.

Requested-drawing gate **PASS**

Correct tag F1 0.5549
Source-shuffled F1 0.0062
No-image F1 0.0000

Correct-minus-controls: +0.5487 / +0.5549

Matched-budget gate **PASS**

3072-side F1 0.5253
768-side F1 0.0003
Common cap 512 tokens

High-minus-low: +0.5250 [0.4224, 0.6293]

Scope gate **BOUNDED**

Qualified
Qwen value-tag candidates within PIDQA
Not qualified
topology, cross-model, or real-plant claims

Qualification is task × model × budget × family

Supported output: human- or downstream-verified candidate-tag retrieval—not general P&ID understanding or autonomous acceptance.